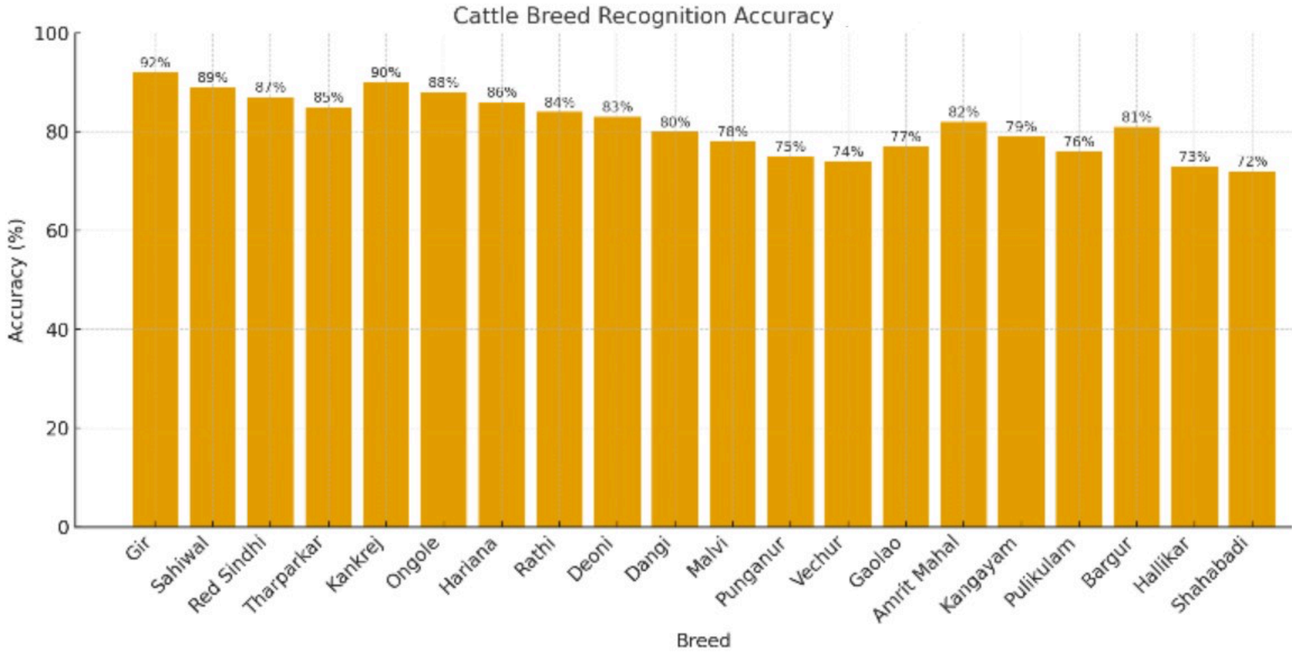
HACKTRIX'25

- **Problem Statement Title-** Image Based breed recognition and animal classification
- **Theme-** Using AI to Identify cattle and buffalo breeds and classify their body types from images
- **PS Category-** Software
- **Team Name:** Verisimilar

AUTOMATED LIVESTOCK EVALUATION AND BREEDING SYSTEM

Proposed Solution

- An AI-based system that uses images to automatically identify the breed of cattle and buffaloes and classify their body traits.
- It minimizes manual errors, ensures consistent and reliable data, and supports better decision-making in livestock programs.
- Combines breed recognition and body type classification in a single, mobile-friendly tool.

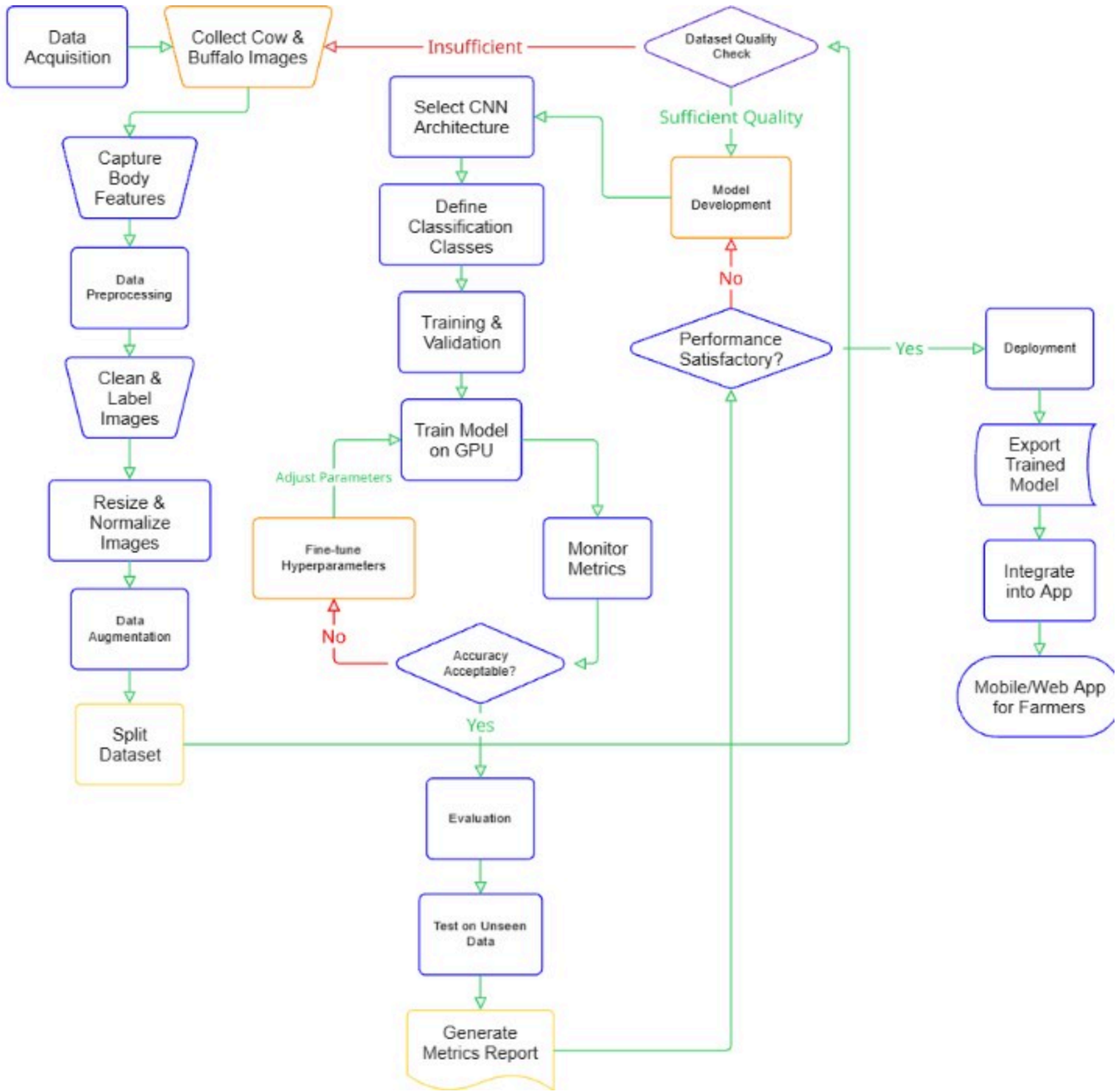


Accuracy (%) of recognizing different cattle breeds using an AI-based breed recognition system.

Done using labelImg and Jtheta

TECHNICAL APPROACH

- **Programming Languages:** Python, Dart (Flutter)
- **Frameworks & Libraries:** TensorFlow, PyTorch, OpenCV, Scikit-learn, XGBoost, SeaBorn, Matplotlib, Google Colab
- **Backend**: Flask (FastAPI)
- **Frontend:** Flutter
- **Database:** PostgreSQL, MySQL
- **Hardware:** Smartphones, Laptop/PC with GPU
- **Machine Learning:** LabellMg, LabelStudio, CVAT, Jtheta



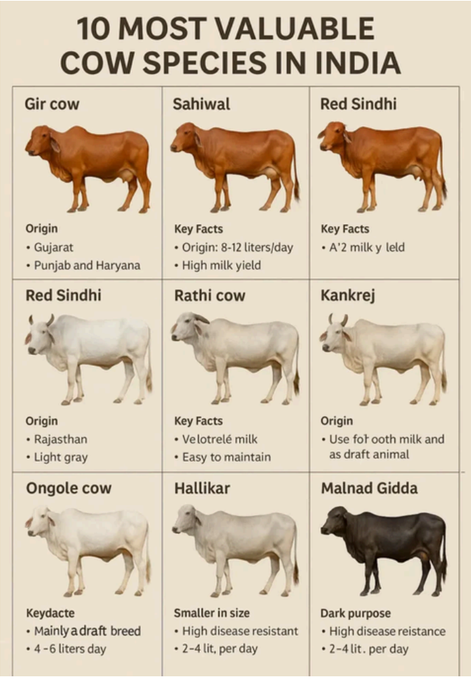
FEASIBILITY AND VIABILITY

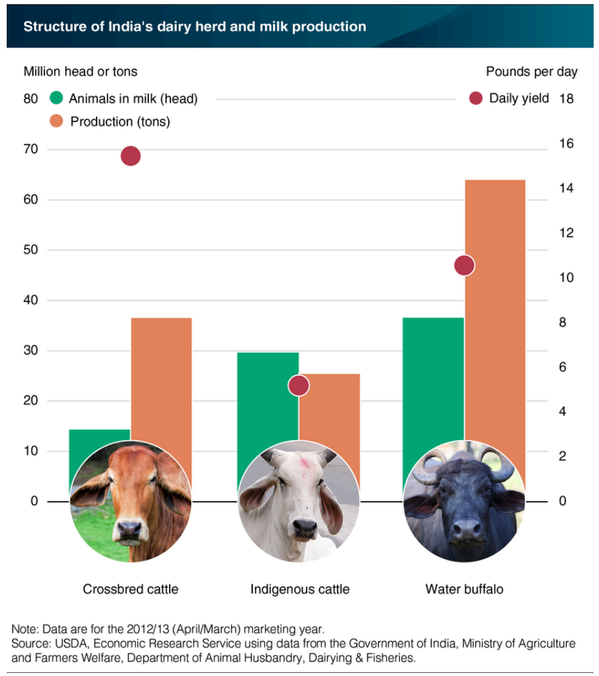
FEASIBILITY	CHALLENGES	SOLUTIONS
Can use smartphones in addition to wildlife surveillance cameras, drones to take photos of animals.	Poor photo quality (blurry or dark images).	Give simple instructions to take clear photos.
AI can detect body parts from images.	Animals may look very similar (crossbreeds).	Train AI on many breeds and crossbreeds.
Scores can be automatically calculated.	AI might make mistakes in unusual poses or angles.	Show a confidence score and let a human verify if needed.
Data can be stored in existing app database.	Field workers may be hesitant to use new technology.	Make the app very simple and easy to use.
Cheaper than manual repeated inspections.	Network issues in rural areas may delay uploads.	Allow offline storage and later syncing.
Can give faster results than humans.	Animals moving or standing awkwardly.	Take multiple photos and pick the best one automatically.

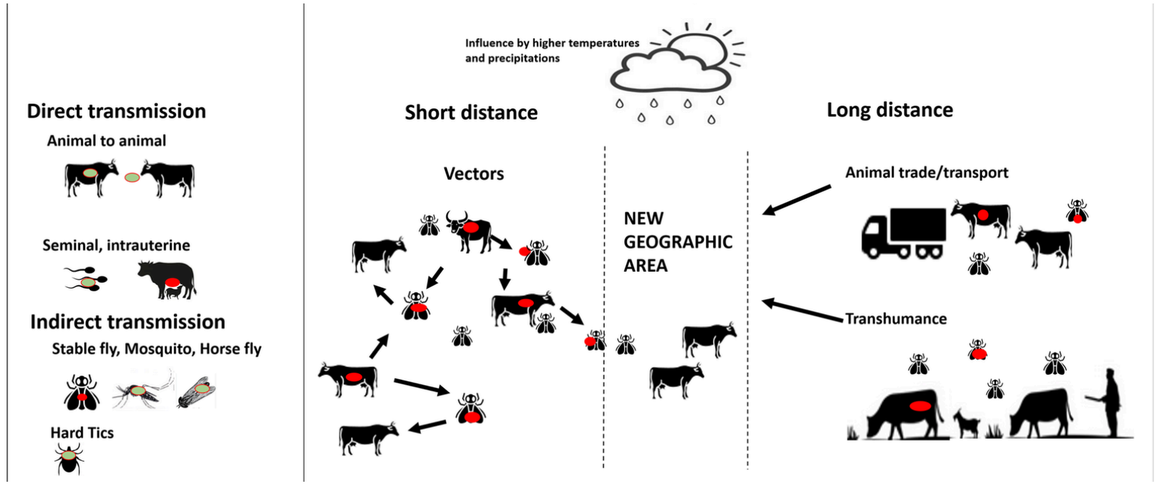
IMPACT AND BENEFITS

IMPACT
Makes evaluating animals faster and easier for field workers.
Reduces mistakes and personal bias in scoring.
Helps identify the healthiest and most productive animals for breeding.
Provides reliable data that farmers and researchers can trust.
Brings simple technology to help improve dairy farming practices.

BENEFITS
Saves time and energy for workers who don't have to measure everything manually.
Ensures fair and consistent scores for every animal.
Easy for anyone to use, even without technical skills.
Less stressful and more cost-effective than traditional methods.
Helps farmers make better decisions about breeding and milk production.



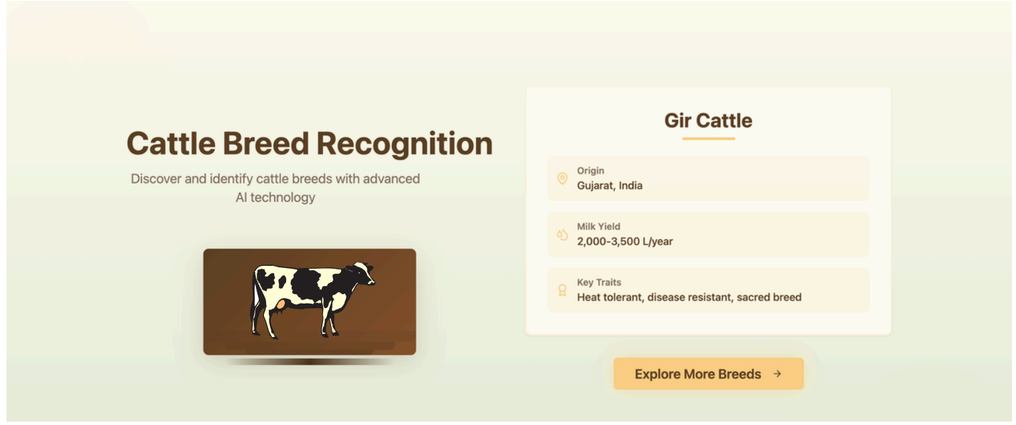


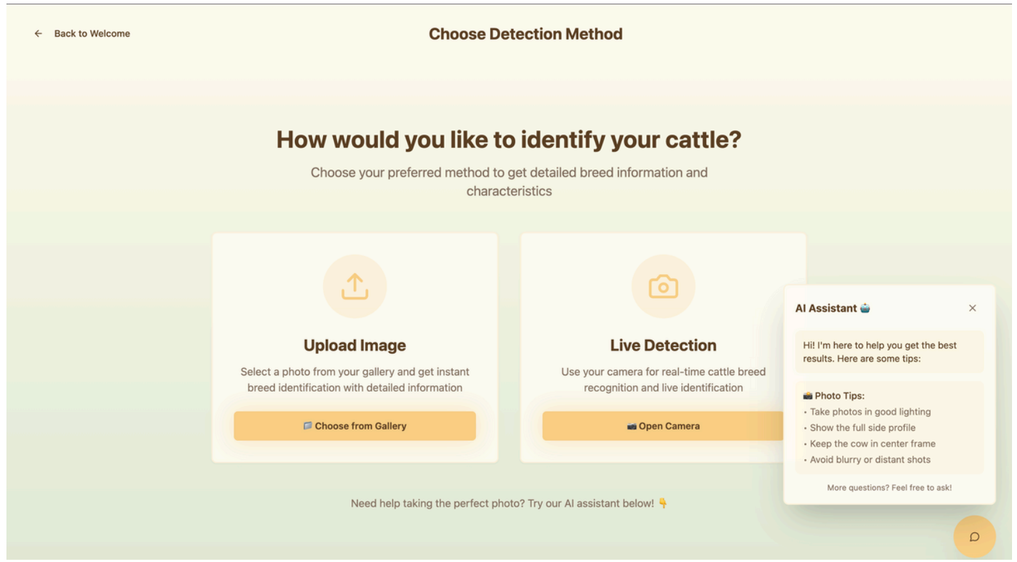


RESEARCH AND REFERENCES

- http://www.agritech.tnau.ac.in/expert_system/cattlebuffalo/Breeds%20of%20cattle%20&%20baffalo.html
- https://hisar.gov.in/cirb/
- ICAR-NIVEDI-PDF: https://nivedi.res.in/Nadres_v2/pdf/covid/BUFFALO%20%26%20CATTLE%20BREEDS.pdf'
- https://www.data.gov.in/keywords/Breed

PROTOTYPE





This is Not the Final Working Prototype